



Note:

- During the attendance check a sticker containing a unique code will be put on this exam.
- This code contains a unique number that associates this exam with your registration number.
- This number is printed both next to the code and to the signature field in the attendance check list.

Maschinelles Lernen

Exam: IN2064 / Retake
Examiner: Prof. Günnemann

Date: Tuesday 11th April, 2023
Time: 16:30 – 18:30

	P 1	P 2	P 3	P 4	P 5	P 6	P 7	P 8	P 9
I									

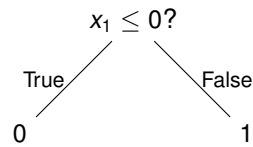
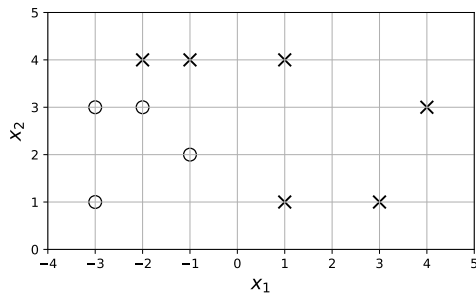
Working instructions

- This exam consists of **12 pages** with a total of **9 problems**.
Please make sure now that you received a complete copy of the exam.
- The total amount of achievable credits in this exam is 41 credits.
- Detaching pages from the exam is prohibited.
- Allowed resources:
 - Two-sided DIN A4 sheet of handwritten notes (a print of digitally handwritten notes is allowed).
- **No other material (e.g. books, cell phones, calculators) is allowed!**
- Physically turn off all electronic devices, put them into your bag and close the bag.
- There is scratch paper at the end of the exam (after problem 9).
- Write your answers only in the provided solution boxes or the scratch paper.
- If you solve a task on the scratch paper, clearly reference it in the main solution box.
- All sheets (including scratch paper) have to be returned at the end.
- **Only use a black or a blue pen (no pencils, red or greens pens!)**
- **For problems that say “Justify your answer” you only get points if you provide a valid explanation.**
- **For problems that say “Derive” you only get points if you provide a valid mathematical derivation.**
- **For problems that say “Prove” you only get points if you provide a valid mathematical proof.**
- If a problem does not say “Justify your answer”, “Derive” or “Prove”, it is sufficient to only provide the correct answer.

Left room from _____ to _____ / Early submission at _____

Problem 1 Decision trees (5 credits)

Consider the following two-dimensional classification dataset with classes “0” (○ marker) and 1 (× marker) and the corresponding decision tree:



- a) Calculate the Gini index $i_G(t_{\text{root}})$ of the root node, $i_G(t_{x_1 \leq 0})$ of the left node, and $i_G(t_{x_1 > 0})$ of the right node. Simplify your results to a single fraction, i.e. $\frac{a}{b}$.

- $i_G(t_{\text{root}}) = 1 - \left(\frac{4}{10}\right)^2 - \left(\frac{6}{10}\right)^2 = \frac{48}{100}$.
- $i_G(t_{x_1 \leq 0}) = 1 - \left(\frac{4}{6}\right)^2 - \left(\frac{2}{6}\right)^2 = \frac{16}{36}$.
- $i_G(t_{x_1 > 0}) = 1 - \left(\frac{0}{4}\right)^2 - \left(\frac{4}{4}\right)^2 = 0$.

- b) Calculate the improvement Δi_G of the Gini index caused by the first split. Simplify your results to a single fraction, i.e. $\frac{a}{b}$.

$$\Delta i_G \quad (1.1)$$

$$= i_G(t_{\text{root}}) - p(x_1 \leq 0) \cdot i_G(t_{x_1 \leq 0}) - p(x_1 > 0) \cdot i_G(t_{x_1 > 0}) \quad (1.2)$$

$$= \frac{48}{100} - \frac{6}{10} \cdot \frac{16}{36} - \frac{4}{10} \cdot 0 \quad (1.3)$$

$$= \frac{128}{600} \quad (1.4)$$

Problem 2 Probabilistic inference (6 credits)

a) You are given N values $x_1, \dots, x_N$ that were independently sampled from uniform distribution $U(0, b)$ with $b \in [0, 1]$ and density function

$$p(x | b) = \begin{cases} \frac{1}{b} & \text{if } 0 \leq x \leq b \\ 0 & \text{otherwise} \end{cases}$$

Derive the maximum likelihood estimate $b_{\text{MLE}} = \arg\max_{b \in [0, 1]} \prod_{n=1}^N p(x_n | b)$.

The likelihood is

$$\begin{cases} \frac{1}{b^N} & \text{if } b \geq \max_i x_i \\ 0 & \text{otherwise} \end{cases}$$

The function is monotonically decreasing in $[\max_i x_i, 1]$, so we must have $b_{\text{MLE}} = \max_i x_i$.

b) Now assume that b is a random variable with $b \sim \text{Beta}(\alpha, 1)$, i.e.

$$p(b) = \begin{cases} \frac{1}{c} \cdot b^{\alpha-1} & \text{if } 0 \leq b \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

where $c \in \mathbb{R}_+$ is a normalizing constant and $\alpha > 0$.

Derive the maximum a posteriori estimate $b_{\text{MAP}} = \arg\max_{b \in [0, 1]} p(b | x_1, \dots, x_N)$.

Note: You will have to make a case distinction based on the value of α .

We have

$$p(b | x_1, \dots, x_N) \propto \begin{cases} b^{\alpha-1-N} & \text{if } b \geq \max_i x_i \\ 0 & \text{otherwise} \end{cases} \quad (2.1)$$

Depending on the value of N , the function is either monotonically increasing, constant or monotonically decreasing in $[\max_i x_i, 1]$. Therefore, we have

$$b_{\text{MAP}} = \begin{cases} 1 & \text{if } N < \alpha - 1 \\ \text{any } b \in [\max_i x_i, 1] & \text{if } N = \alpha - 1 \\ \max_i x_i & \text{otherwise} \end{cases} \quad (2.2)$$

Problem 3 Linear regression (6 credits)

We want to perform regularized linear regression (without bias) on a dataset with N samples $\mathbf{x}_i \in \mathbb{R}^d$ with corresponding targets y_i (represented compactly as $\mathbf{X} \in \mathbb{R}^{N \times d}$ and $\mathbf{y} \in \mathbb{R}^N$). You assume that your targets stem from a generalized Gaussian distribution with parameters $\alpha, \beta > 0$, i.e.,

$$p(y_i | \mathbf{x}_i, \mathbf{w}) = \frac{\beta}{2\alpha\Gamma(1/\beta)} \exp\left(-\left(\frac{|y_i - \mathbf{x}_i^T \mathbf{w}|}{\alpha}\right)^\beta\right) \quad (3.1)$$

with $\Gamma : \mathbb{R}_+ \mapsto \mathbb{R}_+$ being the Gamma function:

$$\Gamma(x) = \int_0^\infty t^{x-1} \exp(-t) dt. \quad (3.2)$$

a) **Derive** the negative logarithm of the likelihood, i.e., $-\log p(\mathbf{y} | \mathbf{X}, \mathbf{w})$ (up to some additive constant w.r.t. $\mathbf{w}$).

$$-\log p(\mathbf{y} | \mathbf{X}, \mathbf{w}) = -\log \prod_{i=1}^N p(y_i | \mathbf{x}_i, \mathbf{w}) \quad (3.3)$$

$$= -\sum_{i=1}^N \log \frac{\beta}{2\alpha\Gamma(1/\beta)} \exp\left(-\left(\frac{|y_i - \mathbf{x}_i^T \mathbf{w}|}{\alpha}\right)^\beta\right) \quad (3.4)$$

$$= \sum_{i=1}^N \left(\frac{|y_i - \mathbf{x}_i^T \mathbf{w}|}{\alpha}\right)^\beta + \text{const in } \mathbf{w} \quad (3.5)$$

b) To add regularization, you add the logistic distribution as prior on your parameters

$$p(\mathbf{w}) = \prod_{i=1}^d \frac{\exp(-\frac{w_i}{s})}{s(1 + \exp(-\frac{w_i}{s}))^2} \quad (3.6)$$

with parameter $s > 0$. **Derive** the negative logarithm of the posterior, i.e., $-\log p(\mathbf{w} | \mathbf{X}, \mathbf{y})$ (up to some additive constant w.r.t. $\mathbf{w}$).

$$-\log p(\mathbf{w}) = -\sum_{i=1}^d -\frac{w_i}{s} - 2 \log\left(s(1 + \exp(-\frac{w_i}{s}))\right) \quad (3.7)$$

$$= \sum_{i=1}^d \frac{w_i}{s} + 2 \log\left(s(1 + \exp(\frac{w_i}{s}))\right)$$

$$-\log p(\mathbf{w} | \mathbf{X}, \mathbf{y}) = -\log p(\mathbf{y} | \mathbf{X}, \mathbf{w}) - \log p(\mathbf{w}) + \underbrace{\log p(\mathbf{y} | \mathbf{X})}_{\text{const. in } \mathbf{w}} \quad (3.8)$$

$$= \sum_{i=1}^N \left(\frac{|y_i - \mathbf{x}_i^T \mathbf{w}|}{\alpha}\right)^\beta + \sum_{i=1}^d \frac{w_i}{s} + 2 \log\left(s(1 + \exp(\frac{w_i}{s}))\right) + \text{const in } \mathbf{w} \quad (3.9)$$

Problem 4 Classification (4 credits)

You are given a dataset with $N = 4$ samples $\mathbf{x} \in [0, 1] \times \mathbb{R}$ and a naive Bayes classifier with classes $y \in \{1, 2\}$ with the prior

$$p(y = k) = \begin{cases} 0.7 & \text{if } k = 1 \\ 0.3 & \text{if } k = 2 \end{cases} \quad (4.1)$$

and feature distributions

$$p(x_1|y = k) = \begin{cases} \text{Bernoulli}(x_1|1) & \text{if } k = 1 \\ \text{Bernoulli}(x_1|0.7) & \text{if } k = 2 \end{cases}, \quad (4.2)$$

$$p(x_2|y = k) = \begin{cases} U(x_2|0, 5) & \text{if } k = 1 \\ U(x_2|0, 1) & \text{if } k = 2 \end{cases} \quad (4.3)$$

where $\text{Bernoulli}(x|p) = p$ is the Bernoulli distribution and

$$U(x|a, b) = \begin{cases} \frac{1}{b-a} & \text{if } x \in (a, b) \\ 0 & \text{else} \end{cases} \quad (4.4)$$

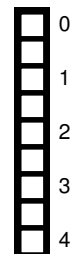
is the uniform distribution.

Classify the following four data points:

	x_1	x_2	x_3	x_4
x_1	1	1	1	0
x_2	0.7	3.5	1.0	0.4

The correct assignments are

y_1	y_2	y_3	y_4
2	1	1	2



Problem 5 Optimization (4 credits)

Below, you are given two different functions and asked to **prove** convexity.

You are given the *quadratic-over-linear* function

$$f(x, y) = \frac{x^2}{y}$$

Prove $f(x, y)$ is convex on the domain $\mathcal{X} = \mathbb{R} \times \mathbb{R}_{>0} = \{(x, y) \in \mathbb{R}^2 | y > 0\}$.

Hint: Remember that a (symmetric) matrix $\mathbf{A} \in \mathbb{R}^{n \times n}$ is positive semi-definite, if $\forall \mathbf{z} \in \mathbb{R}^d : \mathbf{z}^\top \mathbf{A} \mathbf{z} \geq 0$

We show that the second derivative is pos. definite:

$$\nabla^2 f(x, y) = \begin{bmatrix} \frac{2}{y} & -\frac{2x}{y^2} \\ -\frac{2x}{y^2} & \frac{2x^2}{y^3} \end{bmatrix} \quad (5.1)$$

Let $\mathbf{z} = [z_1 \ z_2]^\top \in \mathbb{R}^2$. Then,

$$\mathbf{z}^\top \nabla^2 f(x, y) \mathbf{z} = [z_1 \ z_2] \begin{bmatrix} \frac{2}{y} & -\frac{2x}{y^2} \\ -\frac{2x}{y^2} & \frac{2x^2}{y^3} \end{bmatrix} \begin{bmatrix} z_1 \\ z_2 \end{bmatrix} \quad (5.2)$$

$$= [z_1 \ z_2] \begin{bmatrix} \frac{2z_1}{y} - \frac{2xz_2}{y^2} \\ -\frac{2xz_1}{y^2} + \frac{2x^2 z_2}{y^3} \end{bmatrix} \quad (5.3)$$

$$= \frac{2z_1^2}{y} - \frac{4z_1 z_2 x}{y^2} + \frac{2x^2 z_2^2}{y^3} \quad (5.4)$$

$$= \frac{2}{y^3} [z_1^2 y^2 - 2z_1 y z_2 x + x^2 z_2^2] \quad (5.5)$$

$$= \frac{2}{y^3} [z_1 y - z_2 x]^2 \geq 0 \quad (5.6)$$

The last line inequality follows from $y > 0$.

Because $\mathbf{z}$ is arbitrary, $\nabla^2 f(x, y)$ is pos. semi-definite and hence, f convex.

Problem 6 Deep Learning (4 credits)

Layer normalization improves training stability by normalizing the input to have zero mean and unit standard deviation (similarly to BatchNorm). Layer normalization over the batch dimension takes as input a vector $x \in \mathbb{R}^m$ and transforms it as

$$\mu = \frac{1}{m} \sum_{i=1}^m x_i \quad (6.1)$$

$$\sigma^2 = \frac{1}{m} \sum_{i=1}^m (x_i - \mu)^2 \quad (6.2)$$

$$\hat{x}_i = \frac{x_i - \mu}{\sqrt{\sigma^2 + \epsilon}} \quad (6.3)$$

$$y_i = \gamma \hat{x}_i + \beta \quad (6.4)$$

$$(6.5)$$

In the following tasks, we want to compute the backward pass through this module. As input we get the gradient $\frac{d\mathcal{L}}{dy_i}$ w.r.t. the transformed output.

Derive the gradient of the loss w.r.t to the input x_i , that is, $\frac{d\mathcal{L}}{dx_i}$. First factorize the gradient according to the chain rule and then compute the individual factors.

The factorization of the gradient is given by

$$\frac{d\mathcal{L}}{dx_i} = \sum_{j=1}^m \frac{d\mathcal{L}}{dy_j} \frac{dy_j}{d\hat{x}_i} \frac{d\hat{x}_i}{dx_i}$$

$$\text{with } \frac{d\hat{x}_i}{dx_i} = \frac{d\hat{x}_i}{dx_i} + \frac{d\hat{x}_i}{d\mu} \frac{d\mu}{dx_i} + \frac{d\hat{x}_i}{d\sigma^2} \frac{d\sigma^2}{dx_i}$$

$$\text{and } \frac{d\sigma^2}{dx_i} = \frac{d\sigma^2}{dx_i} + \frac{d\sigma^2}{d\mu} \frac{d\mu}{dx_i}$$

The individual factors:

$$\frac{dy_i}{d\hat{x}_i} = \begin{cases} \gamma, & \text{if } i = j \\ 0 & \end{cases}$$

$$\frac{d\hat{x}_i}{dx_i} = \frac{1}{\sqrt{\sigma^2 + \epsilon}}$$

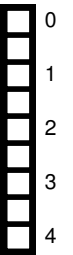
$$\frac{d\hat{x}_j}{d\mu} = -\frac{d\hat{x}_i}{dx_i} = -\frac{1}{\sqrt{\sigma^2 + \epsilon}}$$

$$\frac{d\mu}{dx_j} = \frac{1}{m}$$

$$\frac{d\hat{x}_j}{d\sigma^2} = -\frac{1}{2} (x_j - \mu) (\sigma^2 + \epsilon)^{-\frac{3}{2}}$$

$$\frac{d\sigma^2}{dx_i} = \frac{2}{m} (x_i - \mu)$$

$$\frac{d\sigma^2}{d\mu} = -\frac{2}{m} \sum_{i=1}^m (x_i - \mu)$$



Problem 7 Kernels (6 credits)

- 0 ☐ a) Given vectors a, b and non-negative scalar c and $p \in \mathbb{N}$, **prove** that the polynomial kernel $k(a, b) = (a^T b + c)^p$ is a valid kernel.

1 ☐
2 ☐ **Hint:** You can use the kernel preserving operations from the lecture, given again as reference.
3 ☐ Let $k_1 : \mathcal{X} \times \mathcal{X} \rightarrow \mathbb{R}$ and $k_2 : \mathcal{X} \times \mathcal{X} \rightarrow \mathbb{R}$ be kernels, with $\mathcal{X} \subseteq \mathbb{R}^N$. Then the following functions k are kernels as well:

1. $k(\mathbf{x}_1, \mathbf{x}_2) = k_1(\mathbf{x}_1, \mathbf{x}_2) + k_2(\mathbf{x}_1, \mathbf{x}_2)$
2. $k(\mathbf{x}_1, \mathbf{x}_2) = c \cdot k_1(\mathbf{x}_1, \mathbf{x}_2)$, with $c > 0$
3. $k(\mathbf{x}_1, \mathbf{x}_2) = k_1(\mathbf{x}_1, \mathbf{x}_2) \cdot k_2(\mathbf{x}_1, \mathbf{x}_2)$
4. $k(\mathbf{x}_1, \mathbf{x}_2) = k_3(\phi(\mathbf{x}_1), \phi(\mathbf{x}_2))$, with the kernel k_3 on $\mathcal{X}' \subseteq \mathbb{R}^M$ and $\phi : \mathcal{X} \rightarrow \mathcal{X}'$
5. $k(\mathbf{x}_1, \mathbf{x}_2) = \mathbf{x}_1^T \mathbf{A} \mathbf{x}_2$, with $\mathbf{A} \in \mathbb{R}^N \times \mathbb{R}^N$ symmetric and positive semidefinite

First note that $a^T b$ is a scalar product and hence a kernel. c is a kernel using rule (4) by choosing the feature map $\phi(x) := \sqrt{c}$ and applying the scalar product $\phi(x_1)\phi(x_2)$ as kernel in the new feature space. The sum is kernel preserving (rule 1) as well as a p -times multiplication (rule 3).

- 0 ☐ b) Assume a quadratic kernel $k(a, b) = (a^T b)^2$ with two-dimensional vectors $a, b \in \mathbb{R}^2$ as input. **Derive** a feature map $\phi(a)$ such that $k(a, b) = \phi(a)^T \phi(b)$.

1 ☐
2 ☐
3 ☐

$$(a^T b)^2 = (a_1 b_1 + a_2 b_2)^2 \quad (7.1)$$

$$= a_1^2 b_1^2 + 2a_1 b_1 a_2 b_2 + a_2^2 b_2^2 \quad (7.2)$$

$$= (a_1^2 \quad \sqrt{2}a_1 a_2 \quad a_2^2) \begin{pmatrix} b_1^2 \\ \sqrt{2}b_1 b_2 \\ b_2^2 \end{pmatrix} \quad (7.3)$$

$$= \phi(a)^T \phi(b) \quad (7.4)$$

Therefore, the searched for feature map is $\phi(a) = \begin{pmatrix} a_1^2 \\ \sqrt{2}a_1 a_2 \\ a_2^2 \end{pmatrix}$

Sample Solution

Problem 8 Dimensionality reduction (3 credits)

0 ☐
1 ☐
2 ☐
3 ☐

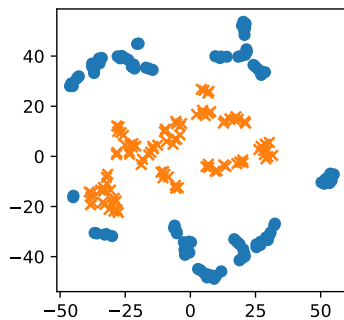
You are given a dataset $\mathbf{X}$ with 200 three-dimensional points. Half of points lie on a sphere with radius 1, and the other half lie on a sphere with radius 0.5, both with a center at origin. You are given routines for PCA as well as TSNE and use them to perform the following transformations:

- a) $\mathbf{A} = \text{PCA}(\mathbf{X}, \text{n_components}=2)$
- b) $\mathbf{B} = \text{TSNE}(\mathbf{X}, \text{n_components}=2, \text{perplexity}=5)$
- c) $\mathbf{C} = \text{TSNE}(\mathbf{X}, \text{n_components}=2, \text{perplexity}=100)$

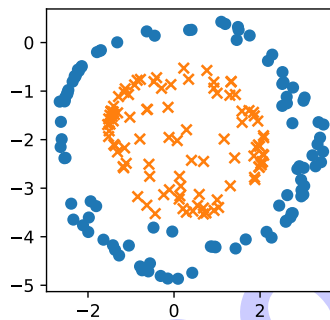
which gives you three down-projected matrices $\mathbf{A}, \mathbf{B}, \mathbf{C} \in \mathbb{R}^{200 \times 2}$.

Three figures below plot the projected data points using a)–c). Blue circles correspond to points which belong to a radius 1 sphere and orange crosses belong to points which belong to radius 0.5 sphere.

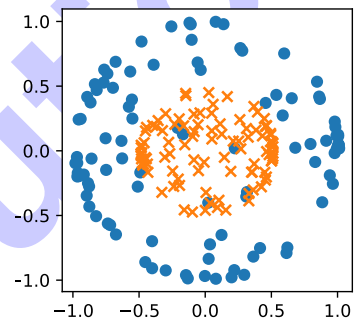
Pair a), b), c) with plots 1), 2), 3) below. Justify your answer.



1)



2)

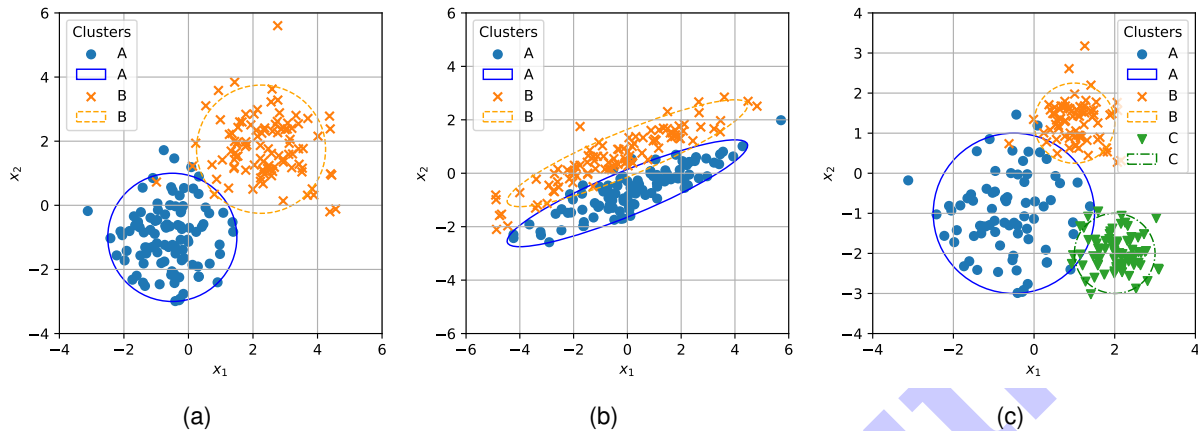


3)

c)–2) because no overlap and large clusters
b)–1) because small clusters
a)–3) because overlap

Problem 9 Clustering (3 credits)

You are given three different two-dimensional datasets $\mathbf{X} \in \mathbb{R}^{n \times 2}$ with varying number of data points n . Dataset (a) and (b) consist of two clusters and (c) of three. The ground truth cluster assignments are given by color and marker style: blue circles, orange crosses, and green triangles (see legend). Each ground-truth cluster follows a multivariate Gaussian distribution. We plot the $4\text{-}\sigma$ covariance ellipses, i.e., each point on this ellipses has equal density (also known as probability contour or likelihood isobar).



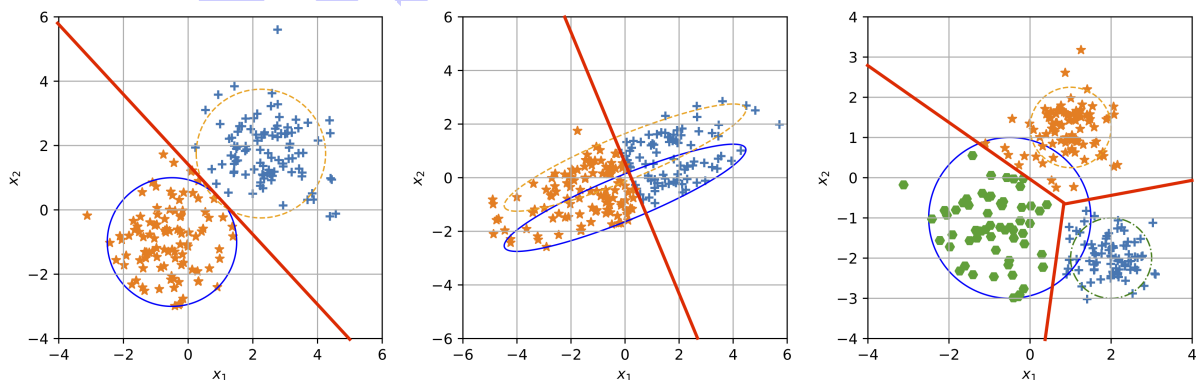
a) What are the optimal decision boundaries for the k-means objective? With decision boundaries we refer to the hyperplanes that are of equal distance to the two closest cluster centroids (also known as Voronoi tessellation). In other words, the points separated by the decision boundaries.

Your task is to **add/sketch the optimal decision boundaries to/in the plots above**. That is, you applied a k-means clustering algorithm that found the globally optimal centroids and you choose k s.t. it matches the number of ground truth clusters.

b) For which dataset is it better to apply k-Means clustering and for which dataset Gaussian mixture models (GMM)? **Justify your answer.**

- (a) K-Means since it has a lower computational cost
- (b) GMM since the covariance matrices contain off-diagonal elements / two spherical clusters would lead to different clusters than the ground truth
- (c) GMM since the covariance matrices differ

K-Means:



Additional space for solutions—clearly mark the (sub)problem your answers are related to and strike out invalid solutions.

A large grid of graph paper, consisting of 30 columns and 30 rows of small squares. A diagonal watermark with the text "Sample Solution" in a light blue, sans-serif font is overlaid across the grid, running from the bottom-left towards the top-right.